
Summer school 2026

Doctoral school: Data Science and AI in Forest ecosystem - atmosphere interactions

- Date: **15. - 22. July 2026**

- Place: **SMEAR Estonia, Järvelja Experimental and Training Forestry Centre**

The summer school will cover topics of Data Science, sensors, IoT, modern technologies in computing on data related to Forest ecosystems and atmospheric measurements.

The event is organised at the Järvelja Training and Experimental Forestry District and the SMEAR Estonia station. We will work in group projects and deploy sensor prototypes, test them on site in the forest at the SMEAR Estonia area.

Learning outcomes are:

- Usage and development of alternative measurement methods to measure environmental parameters in forest ecosystems.
- Learning how to calibrate the measuring tools for environmental data.
- Data processing of the deployed prototypes.
- AI/ML tools to create data processing workflows.

Transferable skills are:

- a. working in a multidisciplinary group as well as making and giving academic presentations
- b. principles of planning scientific experiments
- c. project and time management skills.

Teachers: Steffen M. Noe, Emilio Mercuri, Ahto Kangur, Renato Perotto. The course lecturers and assistants are from the Institute of Forestry and Engineering, Estonian University of Life Sciences, Environmental Engineering Department of the Federal University of Paraná, Brazil, and the Institute of Computer Sciences, Tartu University.